

# An interpretable Dictionary Learning approach for Mult-class abnormality segmentation from 3D Chest CT

# Abstract

Automated Diagnostic Hypotheses from 3D chest Computed Tomography requires clinically interpretable methods to support radiological decision making. Current state-of-the-art (SotA) deep learning models often rely on post-hoc explainability techniques that provide approximate localizations and lack direct traceability to the models decisions. Furthermore, SotA models rarely support simultaneous multi class classification and grounded localization, leading to a fragmented approach to diagnosis. This also hinders the ability to benchmark models across multiple dimensions of performance, including classification accuracy, segmentation quality, and explainability. This study aims to address these challenges through two contributions: First, we introduce an inherently interpretable dictionary learning framework in which detected abnormalities can be backprojected to the original voxel space, allowing diagnostic decisions to be traced back to relevant anatomical structures. Second, we develop a unified evaluation framework to benchmark SotA models across three axes: (i) classification accuracy, (ii) segmentation quality, and (iii) explainability, using standardized metrics and a publically available 3D chest CT dataset with expertly annotated abnormality localizations. Our method is benchmarked against SotA models using this framework, demonstrating competitive classification and segmentation performance while providing improved interpretability compared to deep learning baselines.

# Chapter 1

## Introduction

Developing a unified architectural pipeline for multiclass abnormality classification and grounded localization in 3D Computed Tomography (CT) addresses two of the most critical imperatives in clinical AI adoption: diagnostic accuracy and explainability. Specifically, utilizing dense pixel-level segmentation as the mechanism for localization forces the architecture to provide voxel-level visual evidence for its diagnostic claims. This ensures that the model is held accountable and its reasoning aligns with clinical diagnosis.

The transition from 2D projectional radiography to volumetric CT represents a paradigm shift in medical image analysis. For Lung disease diagnosis, Chest Xrays (CXR) are commonly used as the first imaging test due to its low cost and low radiation exposure. However, studies show that their capability to detect abnormalities is limited when compared to 3D CT. [? ].

Furthermore, CXR has historically benefited from the availability of massive, radiologist-annotated datasets, while the analysis of chest CT volumes has been severely constrained by a lack of such data. However, the recent introduction of large scale, textpaired datasets has facilitated the development of state-of-the-art (SotA) vision language foundation models tailored specifically for volumetric imaging [? ? ? ].

Beyond deep learning approaches, Dictionary Learning algorithms, particularly, Label Consistent KSVD (LCKSVD) [? ], Label Embedded Dictionary Learning (LEDL) and Fischer Discriminant Dictionary Learning (FDDL) offer viable approaches for interpretable CT analysis. All three techniques incorporate label information during the dictionary training process, and ensures that the sparse representation of the data is highly discriminative, making them wellsuited for multiabnormality tasks.

The clinical utility of Deep Learning based approaches is hindered by a lack of interpretability. While tools like GradCAM provide post hoc explanations to model decisions, these are often insufficient for medical professionals who require a granular understanding of the model’s decision making process. In high stakes environments, a model that produces a diagnosis without transparency in its reasoning is difficult to integrate into clinical workflows, as it lacks the accountability necessary for legal and ethical compliance.

## 1.1 Literature Review

The landscape of 3D CT analysis is currently divided into three primary architectural paradigms:

- Classificationbased models that prioritize diagnostic labels
- Segmentationbased models that prioritize boundary precision
- Unified architectures designed for grounded localization

SotA models are predominantly specialized in either classification or segmentation. However, recent systems, particularly in 2025 and 2026, are moving toward multitask learning or foundation models that can perform both diagnosis and voxellevel segmentation simultaneously.

Table 1 summarizes the key models in each category, their primary task orientation, and their performance on standard benchmarks.

Table 1.1: Summary of major 3D CT analysis paradigms and representative models

| Paradigm              | Representative Models                                    | Primary Task                                    | Typical Strengths / Notes                                                                                                                                 |
|-----------------------|----------------------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Classificationbased   | CTCLIP; Merlin; DINOv3                                   | Imagelevel diagnosis; retrieval; linear probing | Strong imagelevel diagnosis; good zeroshot and transfer performance; efficient inference; limited voxellevel localization.                                |
| Segmentationbased     | nnUNet; SegVol; SAMMed3D                                 | Voxellevel lesion segmentation                  | High boundary precision and dense localization; foundation models aim for universal volumetric segmentation across organs and pathologies.                |
| Unified architectures | BiomedParse; MedImageParse; CitrusV; MTMed3D; MedVista3D | Joint diagnosis, segmentation, and reasoning    | Multitask inference with grounded localization; integrates report understanding with voxel outputs; typically more complex and computationally intensive. |

### 1.1.1 Classification Architectures

Classificationbased models for 3D CT have shifted away from 3D Convolutional Neural Networks (CNNs) such as 3D UNet, 3D ResNet and DenseNet pipelines, toward Vision Transformers (ViTs) and selfsupervised encoders. The benchmark for this paradigm is the CTRATE dataset, which introduced the first public largescale 3D medical imaging dataset with paired textual reports, enabling the development of Contrastive LanguageImage Pretraining for Computed Tomography (CTCLIP).

CTCLIP is a VLFM designed for 3D chest CT analysis. It extends the CLIP framework to volumetric medical imaging by jointly learning representations from CT scans and their corresponding radiology reports. [? ? ] Through contrastive learning, the model aligns image features with textual descriptions, enabling tasks such as zeroshot classification, crossmodal retrieval, and reportguided abnormality localization. [? ] The model is particularly effective in learning semantic relationships between radiological findings and volumetric imaging patterns such as GGO, consolidation, fibrosis, and pulmonary nodules. [? ? ]

Recent benchmarks such as Merlin and DINOv3 have further challenged the necessity of domainspecific pretraining. Merlin is a VLFM exclusively pretrained on abdominal CT scans. Despite it’s training, Merlin has consistently demonstrated exceptional zeroshot performance on chest CT classification tasks, even when it was restricted to a linear probe, significantly outperforming models like CTCLIP, M3FM, and CTFM that were explicitly trained on thoracic data, by an average of 12.3% in Area Under the Receiver Operating Characteristic Curve (AUROC). Specifically, in crossdomain retrieval tasks, Merlin surpassed CTCLIP by 24.7%.

DINOv3, a selfsupervised ViT pretrained on over 1.7 billion natural images, outperform specialized medical models like CTNet and CTCLIP across all metrics on several classification tasks due to it’s ability to capture finegrained textural features.

### 1.1.2 Segmentation Architectures

The prevailing SotA for supervised volumetric segmentation remains variations of the 3D UNet, such as nnUNet, which utilizes an adaptive framework to optimize preprocessing and architecture for specific datasets. However, the field has recently shifted toward foundation models capable of universal segmentation such as SegVol and SAMMed3D. The SegVol model is the first 3D foundation model specifically designed for universal volumetric medical image segmentation. Similarly, SAMMed3D adapts the Segment Anything Model (SAM) for 3D medical data.

Recent evaluations published at the Machine Learning for Healthcare (MLHC) Conference in 2025, which specifically utilized the ReXGroundingCT dataset, revealed that the current SotA textprompted segmentation models that demonstrate nearperfect performance in general computer vision tasks struggle with the nuanced clinical descriptions and illdefined boundaries of chest CT abnormalities.

### 1.1.3 Unified Architectures

Recent models have been designed to integrate abnormality detection, segmentation, and multimodal chainofthought reasoning, enabling pixellevel localization, structured report generation, and physicianlike diagnostic inference in a single framework. BiomedParse, and it’s clouddeployed version MedImageParse, developed by Microsoft, are foundation models designed to parse 3D medical images. Other notable models include CitrusV, MTMed3D and MedVista3D, where each architecture incorporates a multitask learning (MTL) framework [? ? ? ], as summarized in Table ??.

### 1.1.4 Dictionary Learning Approaches

Applications of sparse dictionary learning in medical imaging have demonstrated significant success across diverse diagnostic tasks. In histopathological analysis, Discriminative Feature oriented Dictionary Learning (DFDL) has shown the ability to learn class specific dictionaries that achieve high classification accuracy even with limited training data, a critical advantage when generous training samples are not available [? ]. In neuroimaging, multi feature kernel supervised dictionary learning approaches have demonstrated superior performance in distinguishing Alzheimer’s disease patients from cognitively unimpaired individuals, achieving 98.18% classification accuracy while maintaining fast testing times [? ].

Most relevant to our work, dictionary learning has been successfully applied to lung disease diagnosis from CT images. One approach utilized both frozen dictionary learning and label consistent LCKSVD to classify COVID 19 CT scans, achieving 89% accuracy on the CCCCI dataset [? ]. The framework’s key advantage lies in its interpretability, where the sparse coding output can be directly mapped back to the original input signal domain. This work demonstrates that sparse representation methods offer an effective balance between performance and explainability for lung disease classification tasks, providing clear feature representations while maintaining competitive accuracy. Similarly, sparse representation methods have achieved 95.4% accuracy in classifying diffuse lung disease patterns on HRCT images, effectively handling the high variety and complexity of patterns [? ].

### 1.1.5 Explainability Approaches

GradCAM is one of the most widely used posthoc explanation methods for CNNbased classifiers [? ]. It generates classdiscriminative heatmaps by using gradients flowing into the final convolutional layer, allowing the user to identify image regions that influenced a prediction [? ]. It’s main limitation is that it produces lowresolution saliency maps and does not directly provide precise lesion boundaries or voxellevel localization, which is a major drawback for 3D CT interpretation [? ? ].

SegGradCAM extends this idea by adapting GradCAM to semantic segmentation outputs [? ]. Instead of only highlighting regions that are important for imagelevel classification, it produces pixellevel relevance heatmaps for segmentation predictions [? ]. This is particularly useful for volumetric medical imaging tasks where the target abnormality occupies a sparse region within a large 3D scan.

HiResCAM addresses one of the key weaknesses of GradCAM by producing high-resolution and more faithful localization maps [? ]. This is important in chest CT because many abnormalities, such as small nodules, consolidations, or subtle interstitial changes, may be missed by coarse heatmaps. HiResCAM aims to preserve fine spatial detail, making it more suitable for medical imaging applications where precise localization is required [? ]. However, like other gradientbased methods, it remains a posthoc explanation technique and does not change the internal decision process of the model.

# Chapter 2

## Methodology

We implement a sixstage pipeline for our study, consisting of:

1. Data Acquisition
2. Data Preprocessing
3. Implementation of the LCKSVD Approach
4. Abnormality Classification
5. BackProjecting ClassDiscriminative Atoms and Deriving Explainability through Segmentation Masks
6. Evaluation and Benchmarking

### 2.1 Data Acquisition

Two public chest CT datasets: CTRATE[?] and ReXGroundingCT[?] were used in our work. CTRATE was used as the primary source of CT volumes (normal and abnormal). while ReXGroundingCT was used to obtain voxellevel segmentation masks per abnormality scan together abnormality labels, for each corresponding CT volume. Although CTRATE also contains abnormality labels, RexGroundingCT was chosen for its more clinically relevant set of findings, as well as its detailed voxellevel annotations that are essential for our explainability evaluation.

After an initial review of the thoracic findings available in ReXGroundingCT, a discussion with the supervising radiologist produced a shortlist of the following abnormalities that are clinically significant and have sufficient representation in the datasets:

1. Lung nodule
2. GroundGlass Opacity (GGO)

Both the volumes and their segmentation masks are formatted as NIfTI files. The masks are 3D volumetric segmentations ( $H \times W \times D$ ) that align with the corresponding 3D CT scan, with their shape generally aligned with the scan's dimensions, as [512, 512, Depth]. For scans with multiple findings, masks can be structured as [F, H, W, D], where F represents the number of findings. Within a mask file, pixels are labeled with integer values corresponding to different entities for that finding, while 0 represents background.



## 2.2 Data Preprocessing

The data preprocessing pipeline consists of four major steps:

1. Resampling to isotropic spacing
2. Intensity windowing
3. Normalization
4. Patch extraction and vectorization

Each volume is resampled to 1.5 mm isotropic using trilinear interpolation to standardize the spatial resolution across scans. If the current spacing is already within 1% of target, this step is skipped. Each mask is also resampled to exactly match the resampled volume’s spatial shape using nearestneighbour interpolation to preserve integer label values

In the second step, a standard window Hounsfield unit (HU) of [900, 200] was used to separate the lung window from the CT volume. Then, a linear rescaling to [0, 1] as float32 was applied to the windowed volume, to normalize the intensity values.

Finally, the preprocessed volumes are divided into 3D patches of size  $16 \times 16 \times 16$  with 4,096 features per patch ( $24 \text{ mm}^3$  at 1.5 mm spacing). Normal and abnormal scans are processed in separate phases to ensure that the Dictionary Learning model first learns a representation of normal anatomy before being exposed to abnormalities.

In phase 1, patches are extracted on a regular non-overlapping grid with stride of 32 voxels from each normal scan. Only patches with at least 90% lung mask coverage ( $\text{MIN\_LUNG\_COVERAGE} = 0.90$ ) are retained. All patches are labelled with class index 0 (normal).

In phase 2, using the abnormal scans, we slide a 3D patch window over each abnormality’s bounding box with a fixed stride of  $\text{ABNORMAL\_STRIDE} = 2$  voxels. All patches within the bounding box foreground are extracted regardless of specific overlap threshold. Patches with more than 50% zero voxels (background) are discarded. Each patch is directly labeled with the category being processed. For scans with multiple overlapping abnormalities, patches are extracted and labeled separately per category.

Both phases share the same maximum patches/scan budget; class balance is therefore determined by the normal/abnormal scan counts in the dataset, not a hard ratio. All patches are assembled into X ( $4096 \times N$ , float64) and H ( $N$ ,) int64, then saved as .npz files.

## 2.3 Implementation of the LCKSVD Approach

To implement the Dictionary Learning approach for our study, we required a concrete implementation of the LCKSVD algorithm. [?] Currently, many of the available standard implementations of KSVD and LCKSVD are developed using MATLAB, which is not ideal for integration with our Pythonbased data processing and evaluation pipeline. Although A few opensource Python implementations LCKSVD are available, they either lacked an importable library, or did not provide the flexibility required for our study.

Therefore, we developed a custom implementation of the LCKSVD algorithm in Python and published it on PyPI, the official thirdparty software repository for Python, under the name “reppi.” The implementation follows a modular design, consisting of four separate components:

1. SRC Utility Functions shared across Sparse Coding Algorithms
2. OMP Implementation of BatchOMP and OMPCholesky approaches for Sparse Coding
3. KSVD Implementation of KSVD Dictionary Learning Algorithm
4. LCKSVD Implementation of LCKSVD1 and LCKSVD2 Dictionary Learning Algorithms

This design also allows for flexibility in integrating alternative sparse coding methods in the future. The source code is available at [github.com/ckekula/reppi](https://github.com/ckekula/reppi).

This modular architecture enables straightforward integration of the algorithm through library imports, while providing the flexibility required to experiment with different hyperparameters and configurations. The implementations of the SRC, OMP, and LCKSVD modules are provided in Appendix A, Appendix B, and Appendix C, respectively.

## 2.4 Model Training and Inference

Prior to training, each patch vector is L2normalised to unit norm, ensuring that the sparse coding step operates on texture direction rather than absolute intensity magnitude. Patches with a norm below  $1 \times 10^{10}$ , corresponding to uniform zero regions with no signal, are discarded entirely.

The model is trained using LCKSVD2, a dictionary learning algorithm that jointly optimises three objectives: reconstruction accuracy, label consistency, and linear classification. Training proceeds in two stages: 15-iteration KSVD warmstart that initialises the dictionary on reconstruction error alone, followed by 30 LCKSVD2 iterations that refine it under the combined objective. This twostage procedure avoids poor local minima that arise when the labelconsistency and classifier terms are applied before the dictionary has converged to a geometrically meaningful initialisation.

The key hyperparameters are a dictionary size of  $K = 128$  atoms, a sparsity constraint of  $T = 10$  nonzero coefficients per patch (approximately 8% of  $K$ ), a label-consistency weight of  $\alpha = 4.0$ , and a classifier weight of  $\beta = 2.0$ . On smaller datasets where the patch count  $N$  is insufficient to support 128 atoms, the dictionary size is adaptively capped at  $\max(8, N/2)$  and the sparsity constraint is scaled proportionally.

In practice, discriminative atoms are identified by selecting those atoms  $d_k$  whose classifier weight  $|W[c, k]|$  exceeds the 75th percentile of all weights for class  $c$ . This filtering removes reconstruction-focused atoms that lack class-specific discriminative power, focusing the backprojection on class-relevant features.

At inference time, a new CT volume is loaded and preprocessed through the identical pipeline used during training. However, unlike training, instead of randomly sampling patches as in training, inference requires spatial exhaustiveness so that every region of the volume receives a classification score.

This is achieved by extracting patches on a dense regular grid with a stride of 8 voxels along each spatial axis (stride = PATCH\_SIZE / 2 = 16 / 2 = 8). For a volume of size  $512^3$ , this produces approximately  $64^3 = 262144$  patches, each a  $16^3$  voxel window centred on a grid point. The overlap between adjacent patches (stride=8, patch\_size=16) provides spatial coverage with 50% overlap, providing multiple independent score estimates per spatial location for the subsequent backprojection step.

## 2.5 BackProjecting ClassDiscriminative Atoms

The unique intuition we propose in our approach is to generate spatial contribution maps through backprojection. Because each atom in the dictionary  $D$  is associated with a specific class label through the matrix  $Q$  [? ], the model can theoretically "explain" its decision for a test patch by examining which atoms were used in its sparse representation.

To generate a spatial contribution map for a class  $c$ , the algorithm identifies all atoms  $d_k$  that have been assigned to that class. For a query voxel or patch with sparse coefficients  $x$ , the contribution  $C_c$  is calculated as the partial reconstruction:

$$C_c = \sum_{k \in \text{Class } c} d_k x_k$$

This process backprojects the class-specific features into the original voxel space, creating a map that highlights the regions contributing to the detection of a specific abnormality. For classdiscriminative atoms operating on raw voxelspace inputs, the backprojection is an exact, closedform spatial attribution.

## 2.6 Deriving Explainability through Segmentation Masks

To evaluate the clinical accuracy of LCKSVD, these contribution maps are converted into binary segmentation masks for comparison with ground truth annotations. This conversion can be achieved through various thresholding methods.

GradCAM's approximation has two sources of imprecision — the gradient itself is a local linear approximation of a nonlinear function, and the spatial upsampling from the coarse feature map to input resolution introduces further interpolation error. LCKSVD's contribution map has only one source of imprecision: the thresholding step you apply to binarize it into a mask. The map itself — before thresholding — is exact given the sparse code and the dictionary.

Unlike the passive GradCAM approach, atom backprojection is generative. The atoms are what the model actually used to reconstruct and classify the signal. There's no proxy involved. The localization evidence is produced by the same computation that produced the classification, not by a separate posthoc analysis run afterwards.

# Chapter 3

## Results

### 3.1 Evaluation Framework Design

Comparing a sparse linear approach with stateoftheart nonlinear model architectures using only raw performance scores can give an incomplete view of model behavior. In medical imaging, a useful model should not only predict the correct abnormality, but also identify the relevant anatomical region and provide a meaningful explanation for its decision. Therefore, the evaluation framework in this study is organized around three complementary evaluation axes:

- **Axis A: Classification Accuracy** evaluates the ability of the model to detect the presence or absence of each abnormality using standard discriminative metrics such as AUROC, precision, recall, and F1 score.
- **Axis B: Grounded Localization Quality** evaluates whether the model localizes the abnormality in the correct region of the 3D CT volume using spatial overlap and hitbased localization metrics such as DSC, IoU, and HIT@K.
- **Axis C: Interpretability and Explanation Faithfulness** evaluates whether the model’s explanation or attribution map is spatially aligned with the true abnormality region using metrics such as AttributionMask IoU, Pointing Game Accuracy, and Energy Inside Mask.

Axis C is particularly important for this study because the proposed LCKSVD-based approach is designed to provide interpretable decisionmaking through dictionary atoms and sparse activations. In contrast, most deep learning and foundation models require posthoc attribution techniques such as GradCAM, HiResCAM, or saliencybased methods to explain their predictions. These posthoc methods provide useful approximations, but they do not directly represent the internal decision process of the model. Therefore, the evaluation framework explicitly separates predictive performance, localization quality, and explanation faithfulness to better highlight the strengths and limitations of each model type.

### 3.2 Evaluation Metrics

A common set of quantitative metrics was used to evaluate the models across the three axes. This ensures that different models are compared under the same evaluation

criteria, even though their raw outputs may differ. Classification-oriented models are evaluated using disease probability scores, localization-oriented models are evaluated using predicted masks, and interpretable models are additionally evaluated using attribution or activation maps.

### 3.2.1 Classification Metrics

For Axis A, the classification task measures whether the model correctly detects each abnormality. The following metrics are used:

- **Area Under the Receiver Operating Characteristic Curve (AUROC):** AUROC measures how well the model separates positive and negative cases across different decision thresholds. It is useful for imbalanced medical datasets because it evaluates ranking performance rather than relying on a single fixed threshold.
- **Precision and Recall:** Precision measures how many predicted positive cases are actually positive, while recall measures how many true positive cases are correctly detected by the model.
- **F1 Score:** F1 score is the harmonic mean of precision and recall. It provides a balanced measure when both false positives and false negatives are important.
- **Average Precision:** Average precision summarizes the precision-recall curve and is useful when positive cases are relatively rare.

### 3.2.2 Localization and Grounding Metrics

For Axis B, the localization task measures whether the predicted abnormality region overlaps with the groundtruth lesion mask. This is important because a model may correctly predict the presence of a disease while still focusing on the wrong anatomical area.

- **Dice Similarity Coefficient (DSC):** Dice measures the spatial overlap between the predicted mask and the groundtruth mask. It is defined as:

$$DSC = \frac{2|X \cap Y|}{|X| + |Y|}$$

where  $X$  is the predicted mask and  $Y$  is the groundtruth mask.

- **Intersection over Union (IoU):** IoU, also known as the Jaccard index, measures the ratio between the overlapping region and the total combined region:

$$IoU = \frac{|X \cap Y|}{|X \cup Y|}$$

- **HIT@K:** HIT@K measures the proportion of cases where the model achieves at least a small amount of meaningful overlap with the groundtruth lesion region. In this work, HIT@5, HIT@10, and HIT@25 are used to indicate whether the Dice score exceeds 0.05, 0.10, and 0.25 respectively. This is useful for diffuse abnormalities where exact boundary agreement is difficult.

### 3.2.3 Interpretability Metrics

For Axis C, the interpretability task evaluates whether the model’s explanation is focused on the clinically relevant abnormality region. This is different from localization because the goal is not only to check where the model predicts a lesion, but also to determine whether the model’s decision is supported by the correct anatomical evidence.

- **AttributionMask IoU:** This measures the overlap between the thresholded attribution map and the groundtruth lesion mask. A higher value indicates that the explanation is spatially aligned with the true abnormality.
- **Pointing Game Accuracy:** This checks whether the highest attribution point lies inside the groundtruth lesion region. If the maximum response is inside the lesion mask, the case is counted as a hit.
- **Energy Inside Mask:** This measures the percentage of total attribution energy that lies inside the groundtruth mask. A high value indicates that the model’s explanation is concentrated on the true abnormality region rather than irrelevant background structures.

## 3.3 Classification Accuracy Evaluation

The classification performance was evaluated across four abnormality classes: Lung Nodule, Lung Opacity, Consolidation, and Atelectasis. The proposed LCKSVDbased baseline was compared with three foundation model baselines: Merlin, CTCLIP, and BiomedParse. For LCKSVD, consolidationrelated performance was also used as a proxy reference for atelectasis due to current dataset and label constraints.

### 3.3.1 Overall Classification Performance

Figure ?? shows the macrolevel classification performance across the evaluated models. Merlin achieved the strongest discriminative performance, with a macroAUROC of 0.966. In contrast, CTCLIP and BiomedParse produced lower macroAUROC values of 0.538 and 0.509 respectively on the ReXGroundingCT evaluation setting.

These results suggest that Merlin is the strongest classification baseline in this setup. CTCLIP and BiomedParse show weaker classification performance, which may be due to differences in training objectives, dataset alignment, and the use of zeroshot or proxy classification outputs. BiomedParse is mainly designed for textguided segmentation and grounding, so its classification score should be interpreted as a proxy rather than as the output of a dedicated classifier.

### 3.3.2 PerClass Analysis

The perclass AUROC comparison in Figure ?? provides a more detailed view of model behavior. Merlin maintains strong performance across all evaluated classes. The LCKSVD baseline shows promising results for nonfocal or broader abnormalities, achieving AUROC values of 0.917 for Lung Opacity and 0.806 for Consolidation. It



Figure 3.1: Overall classification performance comparison across models.



Figure 3.2: Perclass AUROC comparison Across Models.



also achieves 0.806 for Atelectasis in the current evaluation setting. However, its performance for Lung Nodule is weaker, with an AUROC of 0.524.

This behavior suggests that LCKSVD may capture broader abnormal tissue patterns more effectively than small focal lesions. Lung nodules are compact and localized, so their detection may require finer spatial resolution, multiscale patch modeling, or more nodulespecific training samples. In contrast, abnormalities such as lung opacity and consolidation may produce more distributed visual patterns that are more compatible with the sparse dictionary representation.

### 3.4 Grounded Localization Evaluation

Axis B evaluates whether the models can identify the correct abnormal region in the volumetric CT scan. This is different from classification because a model can correctly detect the presence of a disease while still localizing the wrong anatomical region. The localization evaluation therefore uses Dice, IoU, and HIT@K metrics to measure spatial agreement between predicted lesion regions and groundtruth annotations.

Images/Chapter 4/Localization\_Overall\_Model\_Comparison.png

Figure 3.3: Overall localization performance comparison for BiomedParse and Merlin.

Figure ?? shows that grounded localization remains challenging for the evaluated models. The Dice, IoU, and HIT@K values indicate limited agreement between predicted lesion regions and reference annotations. This highlights an important gap

between disease recognition and true spatial grounding. For clinical support, it is not sufficient to detect an abnormality; the model should also identify the correct location and extent of the abnormal region.

### 3.4.1 PerClass Localization Performance

Figure ?? shows the perclass Dice comparison. The results indicate that localization performance varies substantially across lesion categories. This variation is likely influenced by lesion morphology, size, contrast, and boundary clarity.

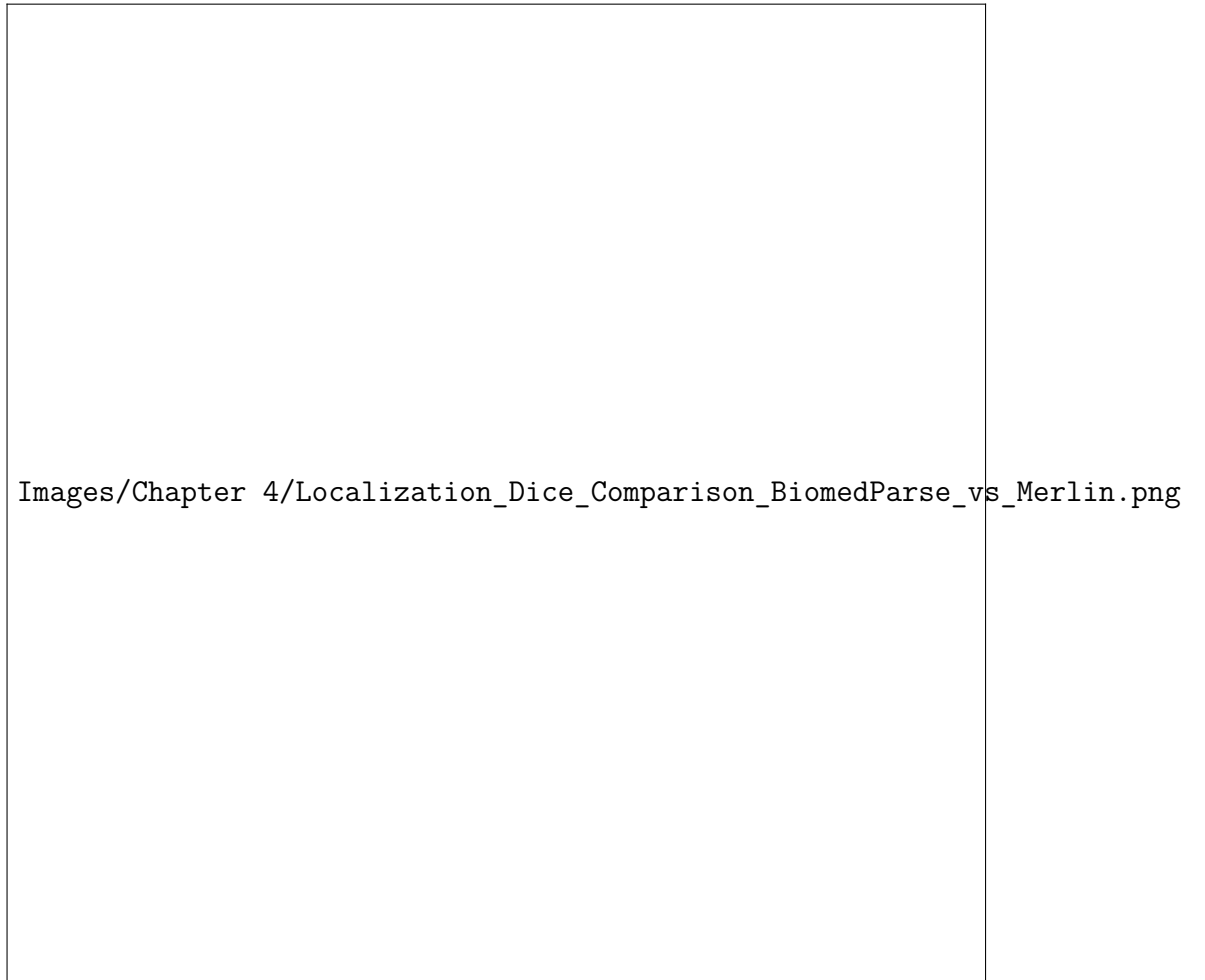


Figure 3.4: Mean Dice comparison by lesion class.

Focal lesions such as lung nodules show comparatively better localization than diffuse abnormalities such as lung opacity, consolidation, and atelectasis. This suggests that objectlike lesions with clearer boundaries are easier for textguided segmentation models to localize. In contrast, diffuse abnormalities often have ambiguous borders and variable appearance, making exact spatial overlap more difficult.

### 3.4.2 Qualitative Comparison of Localization Outcomes

The qualitative BiomedParse example in Figure ?? further illustrates the gap between abnormality detection and precise grounding. In the consolidation example,

the predicted region covers a broad area of the lung, while the groundtruth annotation is more localized. This indicates that the model responds to the general lung region associated with the abnormality, but it does not accurately preserve the lesion boundary, extent, or classspecific spatial location.



Figure 3.5: Groundtruth and BiomedParse-predicted localization for consolidation.

The LCKSVD visualization in Figure ?? shows a related limitation. The prediction map highlights a suspicious region in the lung, indicating some sensitivity to abnormal tissue patterns. However, the activated region is not fully aligned with the groundtruth lesion location. This suggests that the sparse representation captures some abnormal visual structure, but further refinement is needed to improve anatomical alignment.

Overall, the localization results show that both foundation models and the proposed sparse representation approach face challenges in producing precise lesion-level grounding in 3D CT. This supports the need for evaluation beyond classification scores alone.

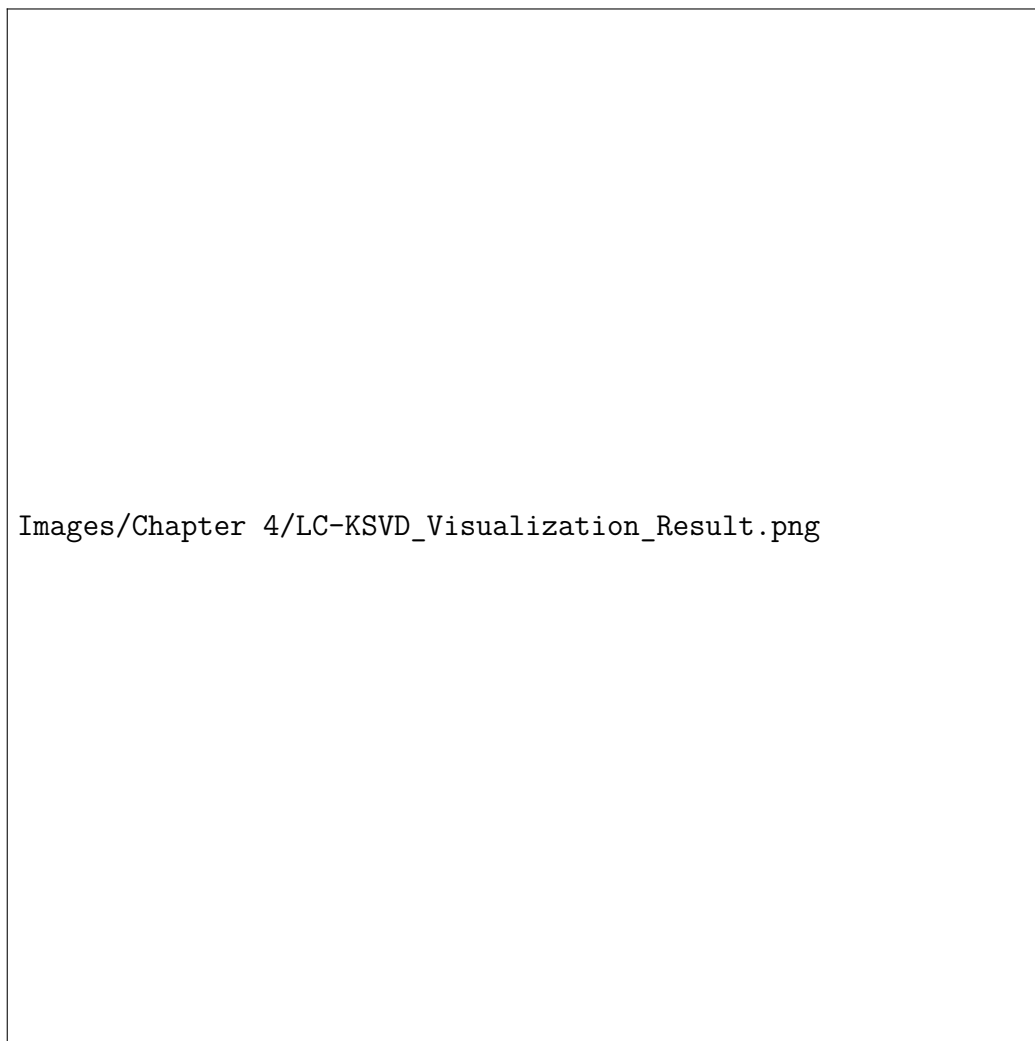


Figure 3.6: Groundtruth and LCKSVDpredicted localization for consolidation.

## 3.5 Inherent Explainability Evaluation

Axis C focuses on whether model predictions are supported by clinically meaningful evidence. This is a key motivation for the proposed LCKSVD approach. Deep learning models such as Merlin, CTCLIP, and BiomedParse can be evaluated with posthoc explanation methods or grounding maps, but these explanations are not always part of the model’s internal decision process. LCKSVD, in contrast, provides a more direct form of traceability through its dictionary atoms and sparse coefficients.

### 3.5.1 AtomtoVoxel Traceability

LCKSVD represents input patches using a learned dictionary and sparse activation coefficients. Since the reconstruction and classification processes are based on linear combinations of dictionary atoms, the contribution of selected atoms can be traced back to the corresponding voxel regions in the CT volume. This provides a direct connection between the model’s prediction and the spatial image patterns that contributed to it.

This atomtovoxel traceability is useful because it allows the model’s decision to be inspected in terms of activated dictionary components. Instead of relying only on a final disease score, the user can examine which atoms were activated and where their contributions appear in the volume.

### 3.5.2 Spatial Faithfulness of Explanations

The explanation quality is evaluated using attribution maps generated from model activations or atom contributions. These maps are compared with the groundtruth lesion masks using the following metrics:

- **AttributionMask IoU:** Measures how much the explanation map overlaps with the true lesion region.
- **Pointing Game Accuracy:** Checks whether the strongest explanation point falls inside the annotated lesion region.
- **Energy Inside Mask:** Measures how much of the explanation signal is concentrated inside the groundtruth mask.

Together, these metrics assess whether the explanation is spatially faithful to the disease region. A model with good interpretability should not only predict the correct abnormality, but should also base its decision on the relevant anatomical evidence.

For LCKSVD, these metrics are particularly meaningful because the explanation comes from the model structure itself through sparse codes and dictionary atoms. Even when its classification performance is lower than that of a large foundation model, its explanations can provide additional value by making the decision process more transparent and easier to inspect.

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